

THEORETICAL ASSIGNMENT

**1D IMAGE BLURRING AND
RESTORATION**

(The Heat Equation)

Topic:

Diffusion of Pixel Intensity Along a One-Dimensional Scan Line

GROUP 12

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Contents

1	Objective	2
2	Theoretical Background	2
3	Problem Formulation and Boundary Conditions	2
4	Separation of Variables	3
5	Solution of the Temporal Equation	3
6	Solution of the Spatial Equation	4
7	Application of Boundary Conditions	4
8	Separated Solutions	5
9	Fourier Series Representation of the Initial Step	5
10	Final Analytical Solution	6
11	Mathematical Proof of Higher-Frequency Decay	6
12	How the Decay Produces Image Blur	6
13	Physical Interpretation	7
14	Final Results	7
15	Conclusion	7

1 Objective

The objective of this assignment is to mathematically study one-dimensional image blurring using the heat equation. A sharp edge in an initially unblurred image is represented by a step function on the interval $[0, L]$. The method of separation of variables is applied to obtain the spatial and temporal solutions. The step function is expanded as a Fourier sine series, and the time-dependent Fourier solution is obtained. Finally, it is proved mathematically that higher spatial-frequency terms decay faster with time, explaining the physical origin of blur.

2 Theoretical Background

In medical imaging and image processing, diffusion of pixel intensity along a one-dimensional scan line can be modeled using the heat equation. Let $I(x, t)$ denote image intensity. The governing equation is

$$\frac{\partial I(x, t)}{\partial t} = \alpha \frac{\partial^2 I(x, t)}{\partial x^2}. \quad (1)$$

Here $\alpha > 0$ is the diffusion coefficient. The temporal derivative describes the rate of intensity change, while the spatial second derivative measures the curvature of the intensity profile.

3 Problem Formulation and Boundary Conditions

Consider $0 \leq x \leq L$. We use homogeneous Dirichlet boundary conditions:

$$I(0, t) = 0, \quad I(L, t) = 0. \quad (2)$$

The initial unblurred image is modeled as a sharp step:

$$I(x, 0) = f(x) = \begin{cases} I_0, & 0 < x < \frac{L}{2}, \\ 0, & \frac{L}{2} < x < L. \end{cases} \quad (3)$$

This represents an ideal sharp edge at $x = L/2$.

4 Separation of Variables

Assume

$$I(x, t) = X(x)T(t). \quad (4)$$

Substitution into Eq. (1) gives

$$X \frac{dT}{dt} = \alpha T \frac{d^2 X}{dx^2}. \quad (5)$$

Dividing by αXT ,

$$\frac{1}{\alpha T} \frac{dT}{dt} = \frac{1}{X} \frac{d^2 X}{dx^2}. \quad (6)$$

Since the left side depends only on t and the right side only on x , both equal a constant $-\lambda$:

$$\frac{1}{\alpha T} \frac{dT}{dt} = -\lambda, \quad (7)$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} = -\lambda. \quad (8)$$

Thus,

$$\frac{dT}{dt} + \alpha \lambda T = 0, \quad (9)$$

$$\frac{d^2 X}{dx^2} + \lambda X = 0. \quad (10)$$

5 Solution of the Temporal Equation

From Eq. (9),

$$\frac{dT}{dt} = -\alpha \lambda T. \quad (11)$$

Hence

$$\frac{dT}{T} = -\alpha \lambda dt. \quad (12)$$

After integration,

$$\ln T = -\alpha \lambda t + C. \quad (13)$$

Therefore,

$$T(t) = C e^{-\alpha \lambda t}. \quad (14)$$

Absorbing C into the spatial coefficient,

$$\boxed{T(t) = e^{-\alpha \lambda t}}. \quad (15)$$

6 Solution of the Spatial Equation

From Eq. (10),

$$\frac{d^2 X}{dx^2} + \lambda X = 0. \quad (16)$$

For $\lambda > 0$, let

$$k^2 = \lambda. \quad (17)$$

Then

$$\frac{d^2 X}{dx^2} + k^2 X = 0. \quad (18)$$

The general solution is

$$X(x) = A \sin(kx) + B \cos(kx). \quad (19)$$

7 Application of Boundary Conditions

From Eq. (2), $X(0) = 0$. Hence

$$0 = A \sin(0) + B \cos(0) = B. \quad (20)$$

so

$$B = 0. \quad (21)$$

Therefore,

$$X(x) = A \sin(kx). \quad (22)$$

The second boundary condition gives

$$0 = A \sin(kL). \quad (23)$$

For a non-trivial solution, $A \neq 0$, hence

$$\sin(kL) = 0. \quad (24)$$

Thus

$$kL = n\pi, \quad n = 1, 2, 3, \dots \quad (25)$$

and

$$\boxed{k_n = \frac{n\pi}{L}}, \quad n = 1, 2, 3, \dots \quad (26)$$

Since $\lambda = k^2$,

$$\boxed{\lambda_n = \left(\frac{n\pi}{L}\right)^2}. \quad (27)$$

8 Separated Solutions

The spatial mode is

$$X_n(x) = A_n \sin\left(\frac{n\pi x}{L}\right), \quad (28)$$

and the time factor is

$$T_n(t) = e^{-\alpha n^2 \pi^2 t / L^2}. \quad (29)$$

Therefore,

$$\boxed{I_n(x, t) = A_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha n^2 \pi^2 t / L^2}}. \quad (30)$$

9 Fourier Series Representation of the Initial Step

The complete solution is

$$I(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha n^2 \pi^2 t / L^2}. \quad (31)$$

At $t = 0$,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right). \quad (32)$$

The Fourier sine coefficient is

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (33)$$

Using Eq. (3),

$$b_n = \frac{2I_0}{L} \int_0^{L/2} \sin\left(\frac{n\pi x}{L}\right) dx. \quad (34)$$

Hence

$$b_n = \frac{2I_0}{n\pi} \left[1 - \cos\left(\frac{n\pi}{2}\right)\right]. \quad (35)$$

Therefore,

$$\boxed{b_n = \frac{2I_0}{n\pi} \left[1 - \cos\left(\frac{n\pi}{2}\right)\right]}. \quad (36)$$

In particular, $b_n = 0$ for $n = 2, 6, 10, \dots$; the remaining non-zero coefficients decrease in magnitude approximately as $1/n$.

10 Final Analytical Solution

Substituting the coefficients into Eq. (31),

$$I(x, t) = \sum_{n=1}^{\infty} \frac{2I_0}{n\pi} \left[1 - \cos\left(\frac{n\pi}{2}\right) \right] \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha n^2 \pi^2 t / L^2}. \quad (37)$$

At $t = 0$ this reproduces the Fourier representation of the sharp step; for $t > 0$ each component is attenuated.

11 Mathematical Proof of Higher-Frequency Decay

The n th mode has the form

$$I_n(x, t) = A_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha n^2 \pi^2 t / L^2}. \quad (38)$$

Its time-dependent amplitude is

$$A_n(t) = A_n(0) e^{-\alpha n^2 \pi^2 t / L^2}. \quad (39)$$

For $m > n$, the ratio of decay factors is

$$\frac{e^{-\alpha m^2 \pi^2 t / L^2}}{e^{-\alpha n^2 \pi^2 t / L^2}} = e^{-\alpha(m^2 - n^2) \pi^2 t / L^2}. \quad (40)$$

For $\alpha > 0$ and $t > 0$, $m^2 - n^2 > 0$, so

$$0 < e^{-\alpha(m^2 - n^2) \pi^2 t / L^2} < 1. \quad (41)$$

Thus the higher-frequency mode decays more strongly. Its decay rate is

$$\gamma_n = \frac{\alpha n^2 \pi^2}{L^2}, \quad (42)$$

so

$$\boxed{\gamma_n \propto n^2}. \quad (43)$$

12 How the Decay Produces Image Blur

A sharp edge contains many Fourier components, including high spatial-frequency terms. These terms represent rapid spatial changes and fine details. Every component is multi-

plied by

$$e^{-\alpha n^2 \pi^2 t / L^2}. \quad (44)$$

Because the exponent contains n^2 , high-frequency components are suppressed much faster than low-frequency components. Consequently, the sharp transition becomes smoother, fine details are lost. Diffusion therefore acts as a low-pass filtering process.

13 Physical Interpretation

The diffusion coefficient α controls the speed of blurring. A larger α produces faster attenuation. From Eq. (42),

$$\gamma_n = \frac{\alpha n^2 \pi^2}{L^2}. \quad (45)$$

Thus smaller spatial scales are smoothed more rapidly. This model is relevant to medical imaging.

14 Final Results

The main results are

$$\boxed{\frac{\partial I}{\partial t} = \alpha \frac{\partial^2 I}{\partial x^2}} \quad (46)$$

$$\boxed{\lambda_n = \left(\frac{n\pi}{L}\right)^2} \quad (47)$$

$$\boxed{b_n = \frac{2I_0}{n\pi} \left[1 - \cos\left(\frac{n\pi}{2}\right)\right]} \quad (48)$$

and the complete solution is Eq. (37). The n^2 dependence of the exponential decay proves that higher spatial frequencies vanish faster, causing the initial sharp image edge to blur.

15 Conclusion

The one-dimensional heat equation has been solved analytically for an initially sharp step-function image intensity profile. Separation of variables produced independent spatial and temporal equations, while the boundary conditions restricted the solution to discrete sine modes. The Fourier coefficients of the step function were calculated explicitly. Most importantly, every spatial mode contains the factor $e^{-\alpha n^2 \pi^2 t / L^2}$. Since the decay rate is proportional to n^2 , higher spatial-frequency components decay faster than

lower-frequency components. High-frequency components carry sharp edges and fine details, so their preferential attenuation smooths the intensity profile and produces image blur. This establishes mathematically how the heat equation models one-dimensional image blurring.

References

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